



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
19-09-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

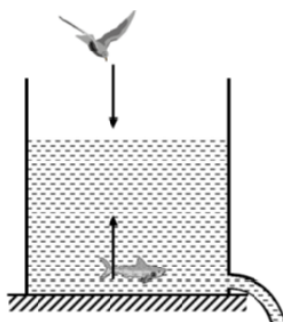
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

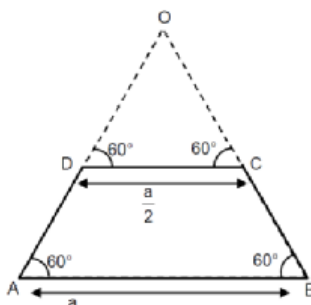
1. A bird in air is diving vertically downward over a tank with speed 5 cm/s while base of tank is silvered. A fish in the tank is rising upward along the same line with speed 2 cm/s. Water level is falling at rate of 2 cm/s. Take $\mu_{\text{water}} = 4/3$.



- (A) Speed of image of fish as seen by bird directly is 6 cm/s.
 (B) Speed of image of fish as seen by bird directly is 3 cm/s.
 (C) Speed of image of fish as seen by bird after reflection from mirror is 3 cm/s.
 (D) Speed of image of fish as seen by bird after reflection from mirror is 6 cm/s.

Space for Rough Work

2. Consider a uniformly charged sheet ABCD, which is a part of an equilateral triangular sheet of side a as shown in figure, $\left(\text{take } \pi = \frac{22}{7}\right)$



Choose the correct option(s) regarding the intensity of electric field at point O due to this sheet.

- (A) Magnitude of $\vec{E}$, increases with the increase in a (keeping charge density same)
- (B) Magnitude of $\vec{E}$, decreases with the increase in a (keeping total charge same)
- (C) If charge density is σ and $a = 1\text{m}$. magnitude of E is equal to $\frac{7\sigma}{44\epsilon_0} \ln\sqrt{2}$
- (D) If charge density is σ and $a = 2\text{m}$ magnitude of E is equal to $\frac{7\sigma}{22\epsilon_0} \ln\sqrt{2}$

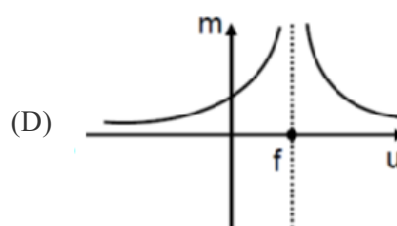
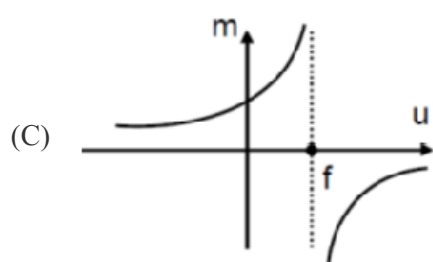
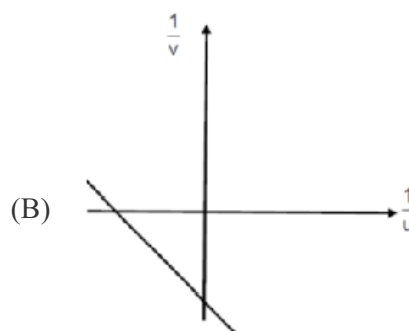
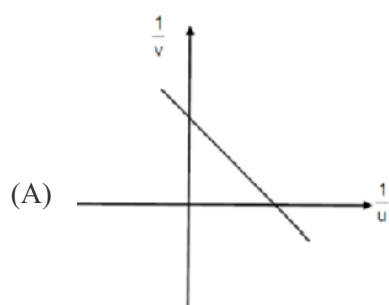
Space for Rough Work

3. Select correct curve for convex mirror:

u = position of object

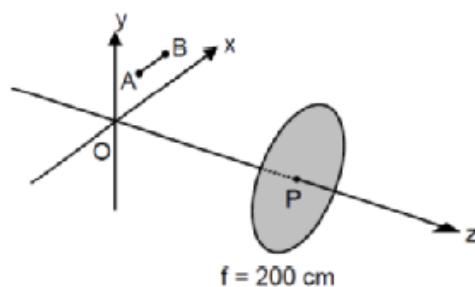
v = position of image

m = transverse magnification



Space for Rough Work

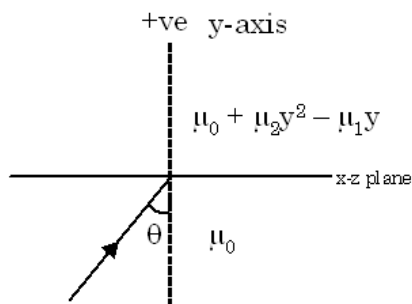
4. A converging lens of focal length 200 cm is placed parallel to x-y plane such that principal axis is along z-axis. The optical centre of the lens is at P (0, 0, 150 cm). Consider two points A(1 cm, 2 cm, 0) and B (2 cm, 2 cm, 0). The image of O, A and B is formed at O', A' and B.



- (A) Coordinates of A' = (4cm, 8cm, 0)
- (B) Coordinates of B' = (8cm, 8cm, - 450 cm)
- (C) $\frac{\text{Area of } \Delta O' A' B'}{\text{Area of } \Delta OAB} = 16$
- (D) The length A'B' = 4cm

Space for Rough Work

5. A medium is separated from other medium of refractive index μ_0 by $x - z$ plane. Refractive index of the medium varies as $\mu = \mu_0 - \mu_1 y + \mu_2 y^2$, choose the correct statement(s). A ray is incident from the other medium to this medium (μ_0, μ_1 & μ_2 are positive)



- (A) Ray will definitely return in the medium of refractive index μ_0
- (B) Ray will never return in the medium of refractive index μ_0 for any value of θ, μ_1 & μ_2
- (C) If ray crosses a particular y , then it will never return to medium having refractive index μ_0 .
- (D) For ray to not return in medium of refractive index $\mu_0, \theta < \sin^{-1} \left[1 - \frac{\mu_1^2}{4\mu_0\mu_2} \right]$

Space for Rough Work

6. A gold leaf experiment is done as shown in fig (i) and (ii) below. A plastic rod was brought close to the top knob the electroscope and taken away. As can be seen in the figures, the angle of divergence between the leaves is seen to increase when the rod is taken away. Which of the following charge combinations can cause this to happen?

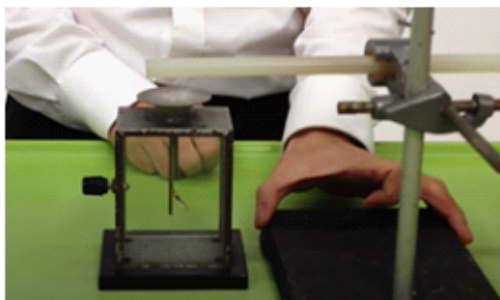


Figure (i)

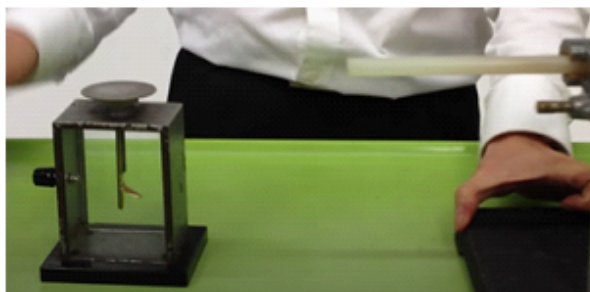


Figure (ii)

- (A) The electroscope as well as the rod are positively charged
- (B) The electroscope is positively charged and the rod is negatively charged
- (C) The electroscope is neutral and the rod is positively charged
- (D) The electroscope is positively charged and the rod is neutral.

Space for Rough Work

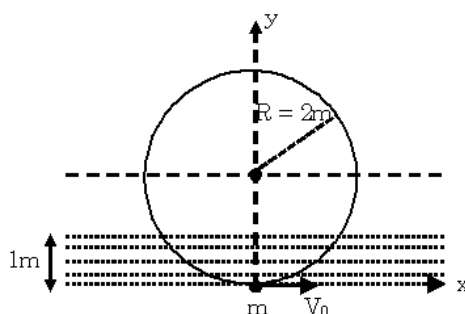
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

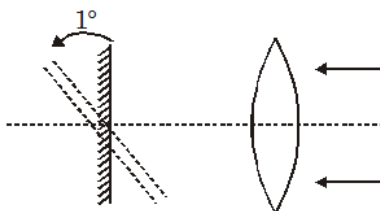
Zero Marks : 0 In all other cases.

1. Consider a uniform electric field $\vec{E} = 3 \times 10^3 \hat{i}$ N/C. What is the flux (in Nm^2/C) of this field through a square of 10 cm side length and normal to the plane makes α, β, γ from x-axis, y-axis and z-axis and $\sin \alpha = \frac{\sqrt{3}}{2}, \sin \beta = \frac{\sqrt{3}}{2}, \sin \gamma = \frac{1}{\sqrt{2}}$
2. A smooth and vertical circular wire frame of radius 2m is fixed inside water as shown. A small bead of specific gravity 0.5 is threaded on the wire and is kept at the origin. If the bead is imparted velocity V_0 towards positive x axis it moves on the wire frame then neglecting effect of viscosity. Minimum value of V_0 so that it complete vertical circle will be (in m/s)



Space for Rough Work

3. A thin equiconvex lens of glass ($\mu = \frac{3}{2}$) having a focal length of 30 cm is placed in air at a distance of 10 cm from a plane mirror, which is placed with its plane perpendicular to the optical axis of the lens. Water ($\mu = \frac{4}{3}$) fills the space between lens and mirror. A parallel beam of light is incident on the lens parallel to the principal axis. IF the mirror is tilted by 1° as shown in the figure, the image shifts from its position by _____ cm.



4. The drag force on a spherical CO_2 bubble in water is given by $F_0 = 6\pi\eta rv$, where η is the viscosity of the surrounding fluid and r is the radius of the sphere (Their relation is called stoke's law). Bubble can be considered to be almost a mass less sphere of radius r moving up in a liquid with terminal speed. Ignore any expansion of bubble during rise. Calculate time taken by sphere to travel distance 20 cm in seconds.
(Given : Radius $r = 3$ mm, density of water $\rho = 10^3$ kg/m³. Density of $\text{CO}_2 = 1$ kg/m³, $g = 10$ m/s², $\eta = 0.5$ SI unit)

Space for Rough Work

5. A spherical mirror is polished on both sides. When the convex side is used as a mirror the image is erect with magnification $\frac{1}{4}$. What is the magnification when the concave side is used as a mirror, the object remaining the same distance from the mirror ?
6. In a setup of displacement method experiment, distance between the screen and a light source is 140 cm and the lens used has a small aperture. By moving the lens between the source and the screen, sharp images are obtained on the screen for two different positions of the lens. The ratio of sizes of these two images is 16 : 1. Find the focal length of the lens (in cm).

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

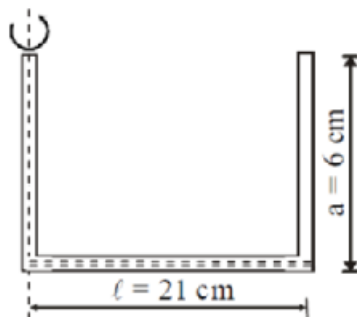
- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

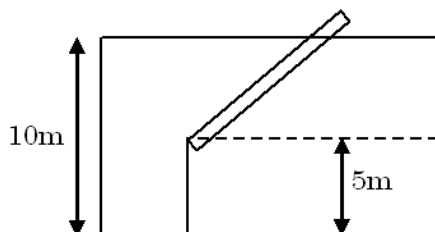
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Length of a horizontal arm of a U-tube is $\ell = 21\text{cm}$ and ends of both of the vertical arms are open to surroundings of pressure 10500 Nm^{-2} . A liquid of density $\rho = 10^3\text{ kg m}^{-3}$ is poured into the tube such that liquid just fills horizontal part of the tube. Now one of the open ends is sealed and the tube is then rotated about a vertical axis passing through the other vertical arm with angular velocity $\omega_0 = 10\text{ radians/sec}$. If length of each vertical arm be $a = 6\text{ cm}$, calculate the length of air column in the sealed arm.

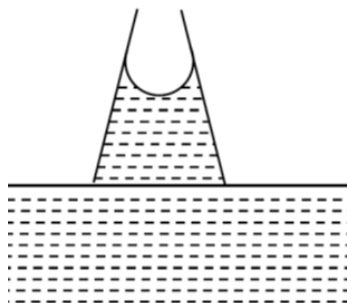


2. A uniform rod of length 6 m has specific gravity $\rho (= 25/36)$. One end of the rod is tied to a 5 m long rope, which in turn is tied to the floor of a pool 10 m deep, as shown. If length of the part of rod which is out of water is $6/x$ meter, find x .

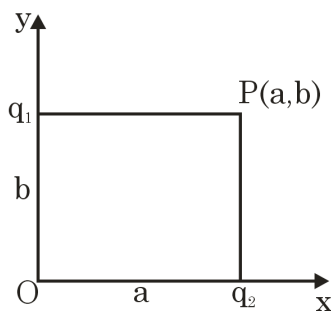


Space for Rough Work

3. Internal diameter of a 10 cm long tapered capillary tube varies uniformly from 1 mm to 0.5 mm from one end to other as shown. The bigger end just touches the surface of a liquid as shown. Surface tension and density of the liquid is $8 \times 10^{-2} \text{ N/m}$ and 1000 kg/m^3 respectively. Angle of contact of the liquid is zero with tube and the meniscus can be assumed hemispheric. Up to what height (in cm) liquid will rise in the tube ? (Take $g = 10 \text{ m/s}^2$)

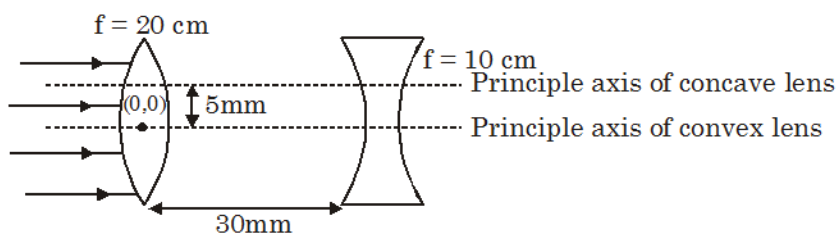


4. Two point charges $q_1 = 2\mu\text{C}$ and $q_2 = 1\mu\text{C}$ are placed at distances $b = 1 \text{ cm}$ and $a = 2 \text{ cm}$ from the origin of the y and x axes as shown in figure. The electric field vector at point $P(a,b)$ will subtend an angle θ with the x-axis given by $\tan\theta = K$. Find value of K.

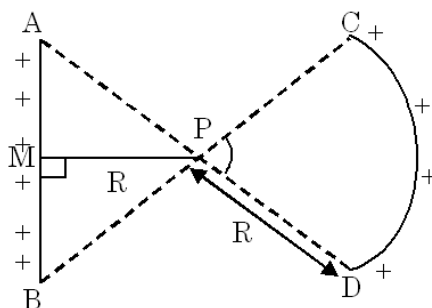


Space for Rough Work

5. If the optical axis of convex and concave lenses are separated by a distance 5mm shown in the figure. If the coordinates of final image formed by the combination of lenses is (x, y) when the parallel beam of light is incident on convex lens. If origin is at the optical centre of convex lens, find $\frac{x}{25y}$.



6. A rod of finite length having linear charge density λ subtends an angle of 15° at a point P which is also the centre of curvature of a circular arc having same linear charge density λ as shown in figure. If the net electric field at the origin is $\frac{bK\lambda}{R}$. Find the value of b:



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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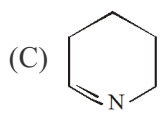
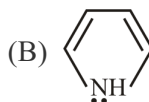
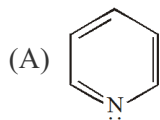
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In which of the following lone-pair is involved in resonance?

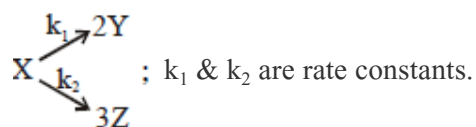


Space for Rough Work

2. In isobutene hyperconjugation involves -

- (A) $\sigma(\text{filled}) - \pi(\text{filled})$ (B) $\sigma(\text{vacant}) - \pi^*(\text{filled})$
 (C) $\pi(\text{filled}) - \pi(\text{vacant})$ (D) $\sigma(\text{filled}) - \pi^*(\text{vacant})$

3. Y and Z are formed in 5 : 3 molar ratio if substance X undergoes 1st order parallel reaction as shown. If activation energy of first reaction is 20 cal/mole and that of second reaction is 15 cal/mole, then



(A) Net activation energy of the reaction is 18.6 cal/mole (approx)

(B) $\frac{k_1}{k_2}$ is $\frac{5}{2}$

(C) Ratio of rate of formation of Y & Z is $\frac{2}{3}$

(D) $k_2 > k_1$

Space for Rough Work

4. If $r_{\text{Na}^+} = 1\text{\AA}$ and $r_{\text{Cl}^-} = 2\text{\AA}$ then the correct statement(s) for NaCl is :
- (A) Distance between nearest cation and anion is $3\text{\AA}$
 (B) Distance between two nearest anion is $4\text{\AA}$
 (C) Edge length of NaCl unit cell is $3\text{\AA}$
 (D) Distance between two nearest Na^+ are $3\sqrt{2}\text{\AA}$
5. In which of the following options order of arrangement does not agree with the variation of property indicated against it?
- (A) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ (increasing ionic size)
 (B) $\text{B} < \text{C} < \text{N} < \text{O}$ (increasing first ionisation enthalpy)
 (C) $\text{I} < \text{Br} < \text{Cl} < \text{F}$ (increasing electron gain enthalpy)
 (D) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ (increasing metallic radius)
6. Find the set of molecules / species having same hybridisation for its central atom and planar :
- (A) BrF_4^- , XeF_4 (B) BH_2^+ , AlBr_3 (C) ICl_2^- , ClF_3 (D) NO_2^+ , N_3^-

Space for Rough Work

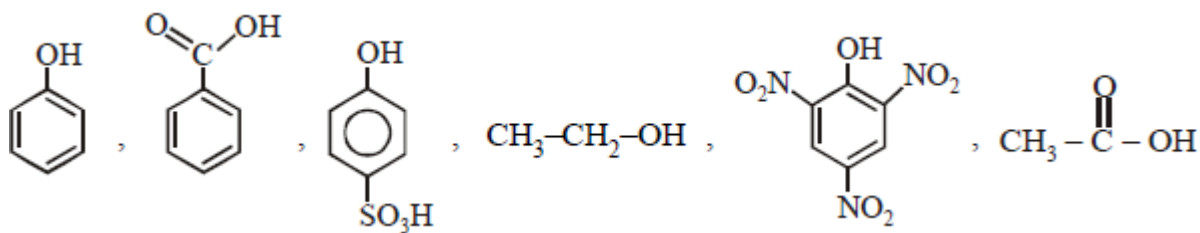
SECTION-II : (Maximum Marks: 24)

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Full Marks : +4 If ONLY the correct numerical value is entered.

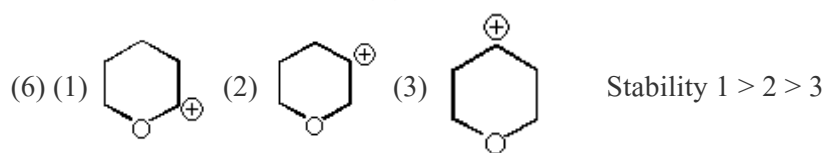
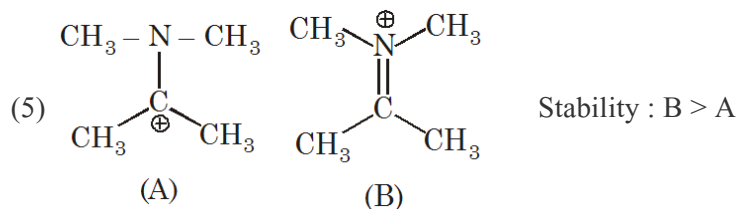
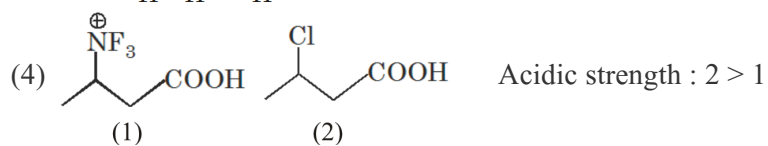
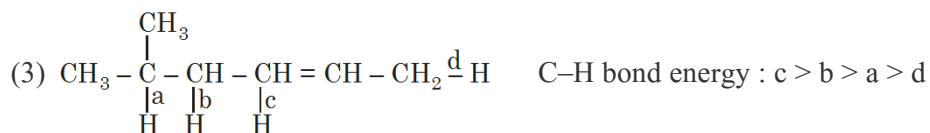
Zero Marks : 0 In all other cases.

1. Calculate the compound(s) which release CO_2 with NaHCO_3 ?



Space for Rough Work

2. How many properties given are **CORRECT** :



Space for Rough Work

3. For any orbital $\Psi(r) = K \cdot e^{-K'r}(r^4 + K_1 r^3 + K_2 r^2)$.
The value of $(n + \ell)$ is :-
4. Find the total number of chemical species which are non planar & lone pair of central atom occupy equatorial orbital.
 SF_4 , XeF_5^- , I_3^- , BrF_5 , XeO_2F_2 , ClF_3 , XeO_3
5. Among the following, total number of compound(s) has/have more boiling point than He.
 H_2O_2 , H_2O , $\text{CH}_3\text{CH}_2\text{OH}$, CH_3OCH_3 , T_2 , D_2 , H_2
6. What volume (in ml) of 0.4 M H_2SO_4 is required to neutralise 1.90 gram of a mixture containing equimolar amounts of NaHCO_3 and Na_2CO_3 ?
(Molar mass : $\text{NaHCO}_3 = 84$, $\text{Na}_2\text{CO}_3 = 106$)
-

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

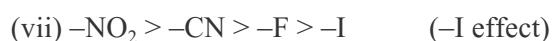
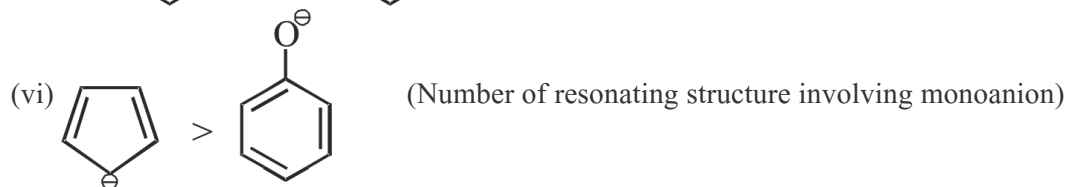
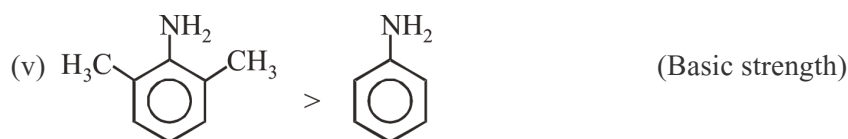
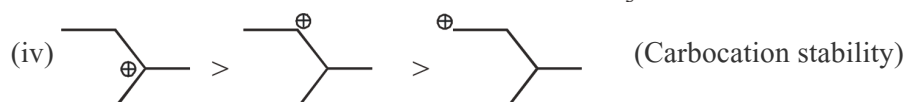
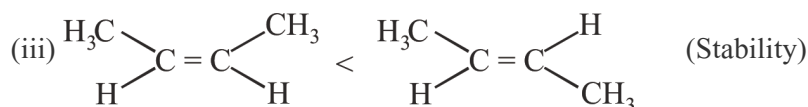
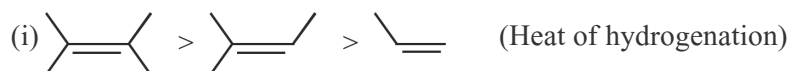
- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

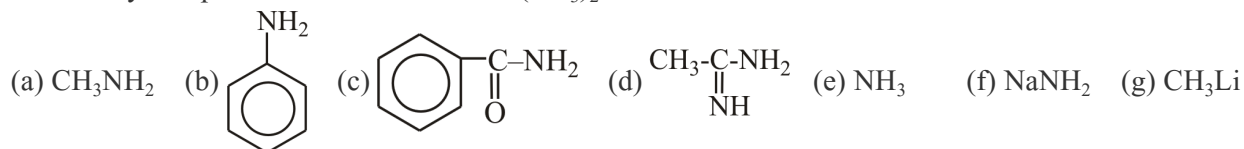
Negative Marks : -1 In all other cases

1. How many of the following orders are correctly matched ?

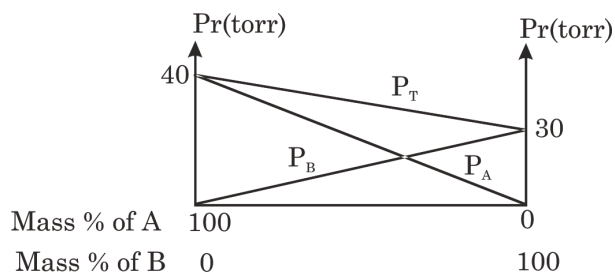


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2. How many compounds are less basic than $(\text{CH}_3)_2\text{NH}$.



3. Equal mass of A & B are present in liquid solution then total pressure exerted by vapours is X torr, then find the value of $\frac{X}{6}$. [A = 80 gm/mol, B = 120 gm/mol]



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4. In the reaction : $\text{N}_2\text{H}_4 (\ell) + \text{ClO}_3^- (\text{aq.}) \rightarrow \text{NO}(\text{g}) + \text{Cl}^- (\text{aq.})$ number of moles of electrons gained by 1.5 moles of $\text{ClO}_3^- (\text{aq.})$ will be :-
5. Number of shortest P – O bonds present in triphosphate ion $\text{P}_3\text{O}_{10}^{5-}$ is.
6. Maximum number of identical type of bond angle(s) in HClO_4 is :
-

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PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If a straight line passes through the point $P(2, 0)$ having equation $y = m(x - 2)$, $m \neq 0$ and intersects the curve $y^2 = 8x$ at the points Q, R (let $PQ = \ell_1$ & $PR = \ell_2$). Also given that G, A represents geometric and arithmetic means of ℓ_1 and ℓ_2 respectively, then correct option(s) is/are

(A) $\ell_1, \ell_2 \geq 2$

(B) $\ell_1, \ell_2 < 2$

(C) $G \propto A$

(D) $G \propto \sqrt{A}$

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2. Let from the point with abscissa 25, two tangents are drawn to the ellipse $24x^2 + 25y^2 = 600$ with foci at S_1 and S_2 . The points of contact of tangents are A and B. If the distance of A from S_1 is $\frac{60}{13}$ units, then
- (A) The distance of A from S_2 is $\frac{80}{13}$ units
- (B) The distance of B from S_1 is 5 units
- (C) The distance of B from S_2 is 5 units
- (D) The distance of A from the directrix corresponding to S_2 is $\frac{350}{13}$ units.
3. The curve $xy = c$ ($c > 0$) and the circle $x^2 + y^2 = 1$ touch each other at two points, then -
- (A) $c = 1/2$
- (B) distance between points of contact is 2 unit
- (C) slope of tangent at point of contact is -1
- (D) length of latus rectum of $xy = c$ is 2

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4. A line divides a plane into 2 regions. Two lines divide the plane into maximum 4 regions. If L_n is the maximum number of regions into which a plane can be divided by n lines then which of the following is/are true :
- (A) $L_{20} = 211$ (B) $L_{10} = 56$ (C) $L_{15} = 121$ (D) $L_{25} = 326$
5. Let $N = 2^4 \times 3^5 \times 5^2 \times 7^4 \times 11$ be resolved as a product of two factors (order immaterial), then
- (A) number of cases in which the two factors are coprime is 16
(B) number of cases in which their H.C.F. is 3, is 32
(C) number of cases in which their H.C.F. is 5, is 16
(D) number of cases in which their H.C.F. is odd, is 180
6. 8 cards and 8 envelopes are numbered 1, 2, 3, 4, 5, 6, 7, 8 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 2 & 4 always placed in envelopes numbered 1 & 3 respectively, then number of ways of placing cards in envelope with above constraint is -
- (A) 255 (B) 362 (C) 9 (D) 97

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. $y + x = 0$ and $y - x = 0$ are tangent at the vertex and axis of a parabola respectively and $y = 2x + 3$ is a tangent to the same parabola and if the length of square of latus rectum is L , then $\frac{9L}{20}$ equals
 2. If N is number of pairs of divisors of 75600 such that product of these divisors is 60480, then the value of $\frac{N}{10}$ is
 3. Following quantities of the given flavours are available in a shop (Ice-creams of the same flavour are identical) :- 4 Vanilla, 5 Butterscotch, 6 Strawberry, 3 Chocolate, 2 Kaju-pista. Then the number of ways in which at least 4 ice-creams can be selected, such that selection of Vanilla is compulsory and Strawberry are taken in pairs or not at all is equal to N then $\frac{N}{100}$ equals
-

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4. For all admissible values of the parameter 'a' the straight line $2ax + y\sqrt{1-a^2} = 1$ will touch an ellipse whose eccentricity is equal to-
5. Let P_n denotes the number of ways of choosing three points out of n distinct points in a line such that no two of them are consecutive. If Q_n denotes the number of ways of choosing similar points on the circle then $\frac{Q_6}{P_6} + \frac{Q_7}{P_7} + \frac{Q_8}{P_8}$ is
6. Let $(ax^2 + by^2 - c)(ax^2 - cy^2 + b) = 0$ represent a hyperbola and an ellipse, where $a, b, c \in \{-3, -2, -1, 1, 2, 3\}$. If N is number of possible ordered triplets (a, b, c), then $\frac{N}{10}$ equals

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Let tangents be drawn to the curve $y^2 - 4(x + y) = 3 \sin \theta + 4 \cos \theta - 15$, $\theta \in \mathbb{R}$ from the origin where the vertex of the curve is at maximum distance from the origin. If slopes of these tangents are m_1 and m_2 , then find the value of $\left| \frac{1}{m_1 m_2} \right|$.
2. Let (x, y) be a variable point on the curve $x^2 + 4y^2 - 4x + 8y + 4 = 0$. Then $\frac{\max(x^2 + y^2 + 4x + 2y + 5)}{\min(x^2 + y^2 + 4x + 2y + 5)}$ is
3. From any point on line $(t + 2)(x + y) = 1$, $t \neq -2$ tangents are drawn to ellipse $4x^2 + 16y^2 = 1$. If chord of contact passes through a fixed point, then number of integral values of t for which fixed point lies inside ellipse, is

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4. Tangents are drawn from any point on straight line $y = x + 8$ to the auxiliary circle of $x^2 + 16y^2 = 16$. If A and B are the points of contact of these tangent and P, Q are corresponding points of A and B on the ellipse respectively. If locus of mid point of PQ is $x^2 + \alpha y^2 + \beta x + \gamma y = 0$ then value of $(\alpha - \beta + \gamma)$ is
5. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. If the locus of the mid-point of the chord of contact is $4(x^4 + y^4 + 2x^2y^2) = \beta x^2 - \gamma y^2$ then the value of $(\gamma - 2\beta)$ is equal to
6. Let there be 15 letters $\left(L, M, N, O, P, \underbrace{Q, Q, Q, \dots Q}_{10 \text{ Q's}} \right)$. If the number of different arrangement of these letters in a line so that there is atleast two alike letters (Q) between two distinct letters is $\frac{k!}{2}$, then 'k' is equal to

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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